Neural networks trained on compact group composition learn the global form of the group

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Abstract

Small neural networks trained to identify products in a compact Lie group, observed only through a generic scrambled encoding, build the irreducible representations of the group's global form: across ten group forms drawn from $\mathrm{SU}(2)$, $\mathrm{SU}(3)$, $\mathrm{Spin}(4)$, and $\mathrm{Spin}(5)$ and their central quotients, the learned code concentrates on a small faithful carrier determined by the quotient, with half-integer spins present exactly when the data distinguish the double cover, deterministically across seeds. Interpretability instruments carry the discovery beyond this correlation. Causal ablations show each isotypic block is necessary for exactly the discriminations its central character supports. A structural test based on the two-sided translation action shows the learned code is the full matrix-coefficient block of its carrier, at the real dimension the real Schur lemma prescribes, and a matched training comparison on $\mathrm{SO}(3)$ shows why: the 9-dimensional block attains a larger margin than a 6-dimensional faithful frame at 17% lower total parameter norm, so regularized training selects the block as the cheaper complete mechanism. Under regression onto heat-kernel targets, the same architecture fills isotypic sectors in order of Casimir eigenvalue, with a truncation frontier set by the weight decay rather than by capacity. The same instruments force one revision: on $\mathrm{SO}(5)$, a stable vector-plus-adjoint mixture survives every replication, so the minimal-carrier description fails as stated in one of the ten group forms.

1 Introduction

A compact Lie group carries a canonical harmonic analysis: by the Peter–Weyl theorem, the matrix coefficients of its irreducible unitary representations span a dense subspace of L^2 [Bröcker and tom Dieck, 1985, Ch. III]. For *finite* groups, mechanistic interpretability has established that small networks trained on the multiplication table represent elements through matrix coefficients of a few irreducible representations [Chughtai et al., 2023, Sec. 4], that competing descriptions of the same models exist [Stander et al., 2024, Sec. 5], and that per-argument equivariance reconciles them Wu et al. [2025]. All of this work presupposes a finite table and one learned embedding per element, the setting of the grokking literature Power et al. [2022], Nanda et al. [2023], Gromov [2023], Zhong et al. [2023], and the associated theory proves irreducible features emerge for finite groups only Morwani et al. [2024], Marchetti et al. [2024], while equivariant architectures place the harmonics into the network by construction Cohen and Welling [2016], Geiger and Smidt [2022]. Whether this representation theory emerges in a generic network when the group is a continuum has remained open.

In this paper, we treat interpretability as a measurement instrument and use it to answer that question and to uncover regularities that were not part of the question. We train encoder-factorized scorers

Submitted to 38th Conference on Neural Information Processing Systems (NeurIPS 2024). Do not distribute.

to identify products in compact groups presented only through frozen generic observation maps, and we analyze the learned representations with isotypic spectroscopy, function-space ablations, activation patching, and a new structural test built on the two-sided translation action of the group on its own matrix coefficients. Four findings result, the last a negative result that revises the first.

The learned carrier identifies the global form of the group. For each central subgroup $Z \subseteq Z(G)$, averaging the observation map over Z realizes the same task on the quotient G/Z, so the global form of the group becomes an experimental variable. Across ten group forms (Table 1), the encoder concentrates 93–98% of its variance on a small faithful carrier of the task group, identically in every seed, with forbidden sectors at the measurement floor. The selection is causal: ablating the carrier collapses ranking to chance and destroys exactly the center discriminations its central character supports, while equal-rank ablations elsewhere are inert (Section 3).

The learned code is the full two-sided coefficient block, selected on cost. The active subspace has exactly the real dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} \mathrm{End}_G(W)$ of the full coefficient block of its carrier W, and fitted left and right translation operators fit at held-out $R^2 \geq 0.98$ and commute, which characterizes the full block within its sector (Lemma 2). A 6-dimensional faithful frame on $\mathrm{SO}(3)$ passes the left test exactly and fails linear right closure, and in matched training the 9-dimensional block attains a larger margin at 17% lower parameter norm (Section 4): optimization selects the block not as the smallest separating code, which it is not, but as the cheapest complete mechanism.

Regression reveals a Casimir-ordered spectral bias with a regularization-set frontier. Trained to regress heat-kernel values $K_t(ab\,c^{-1})$, the same architecture recovers the required sectors at their required weights up to a sharp cutoff that advances in order of Casimir eigenvalue during training, a compact-group form of the spectral bias of Rahaman et al. [2019], and on a band-limited target the in-support sector that weight decay 1.0 prunes is retained at weight decay 0.1 (Section 5).

The instruments force one revision. On $\mathrm{SO}(5)$, a stable vector-plus-adjoint mixture survives five seeds and every conditioning variation we ran, so the minimal-carrier description fails as stated in one of ten group forms, and we report it as such rather than explaining it away (Section 6).

Our methods are standard. What is new is the object, a continuum group with no embedding table, the quotient construction that turns the global form into an experimental variable, and the two-sided translation test of Section 4. The rest of this paper is organized as follows. In Section 2, we fix the task and the instruments. In Sections 3 to 5, we present the carrier, mechanism, and spectral results. In Section 6, we state the revision and a selection conjecture. The background from representation theory and harmonic analysis is collected in Appendix A, proofs are in Appendix B, experimental details in Appendix C, and complete result tables and additional figures in Appendix D.

2 Task, observations, and instruments

Throughout, G denotes a compact Lie group with Haar probability measure, $Z \subseteq Z(G)$ a central subgroup, W an irreducible real representation of G with commutant $\mathrm{End}_G(W) \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, $z(g) \in \mathbb{R}^{128}$ the trained argument-encoder activations, and χ_ρ the character of an irreducible unitary representation ρ.

Task. Given encodings of $a, b \in G$ and of $K = 64$ candidates, the model must identify the product. The positive candidate is $(ab)\exp(\xi)$ with ξ a centered Gaussian of scale $\sigma = 0.2$ on the Lie algebra, the $K - 1$ negatives are Haar samples excluded from a ball around the positive class, and the loss is cross-entropy over the K scores. The Bayes-optimal rule ranks candidates by distance to ab and attains accuracy 0.951 under the $\mathrm{SU}(2)$ sampling. Training pairs come from a pool of 2048 Haar elements, and we report ranking accuracy on pairs of fresh elements never seen in training.

Observations. Each element is presented through a frozen map Ψ rather than a token, for $\mathrm{SU}(2)$ the map $\Psi(q)_i = \exp(\alpha\langle q, h_i\rangle)$ with 32 fixed Haar landmarks h_i and $\alpha = 3$, standardized under Haar statistics. The quotient condition for G/Z replaces Ψ by its average over the Z-orbit, which is exactly Z-invariant, so the observations determine elements of G/Z and nothing finer, with all else shared between conditions. Every result below reproduces under two further observation maps (Appendix C), so the selected carrier is a property of the task and the group rather than of the map.

Model. An encoder-factorized scorer with about 0.3M parameters: a shared two-layer GELU encoder produces $z(a), z(b)$, an interaction network produces $h(a, b) \in \mathbb{R}^{128}$, a candidate encoder

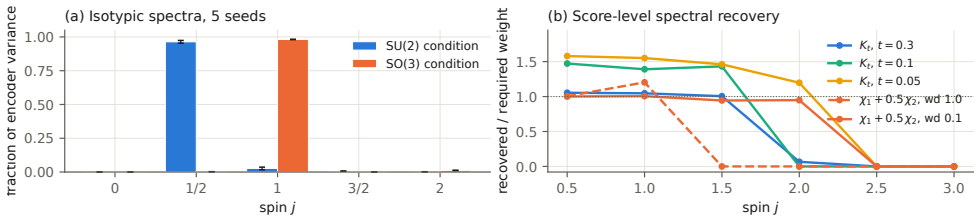


Figure 1: (a) Isotypic spectra of the trained encoder for the $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ conditions (five seeds). (b) Score-level spectral recovery for heat-kernel and band-limited targets. Values near 1 indicate a sector carried at its required weight, and the dashed curve shows the in-support spin-2 sector pruned at weight decay 1.0.

Table 1: Learned carriers across ten group forms, with the fraction of encoder variance on the carrier (mean over five seeds) and the rejected faithful alternative where one exists. $\mathrm{Spin}(4)$ sectors are pairs (j_L, j_R). Seed statistics and causal results per condition are in Appendix D.

task group	learned carrier	variance	rejected alternative
$\mathrm{SU}(2)$	spin $\frac{1}{2}$	0.96	
$\mathrm{SO}(3)$	spin 1	0.98	
$\mathrm{SU}(3)$	$\mathbf{3} \oplus \bar{\mathbf{3}}$	0.97	
$\mathrm{PSU}(3)$	adjoint $\mathbf{8}$	0.93	
$\mathrm{Spin}(4)$	$(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$	0.98	(no faithful irrep)
$\mathrm{SO}(4)$	$(\frac{1}{2}, \frac{1}{2})$	0.97	
$\mathrm{SO}(3) \times \mathrm{SU}(2)$	$(1, 0) \oplus (0, \frac{1}{2})$, 5 dims	0.98	$(1, \frac{1}{2})$, 6 dims: 0.008
$\mathrm{PSO}(4)$	$(1, 0) \oplus (0, 1)$, 6 dims	0.98	$(1, 1)$, 9 dims: 0.011
$\mathrm{Spin}(5)$	spinor $\mathbf{4}$	0.97	
$\mathrm{SO}(5)$	vector $\mathbf{5}$ + adjoint $\mathbf{10}$	$0.69 + 0.22$	(see Section 6)

produces $z_c(c)$, and the score is $\langle h(a,b), z_c(c) \rangle$, trained by AdamW with weight decay 1.0 unless stated. Every reported quantity is a mean over five seeds with its standard deviation.

Instruments. Isotypic spectroscopy computes, on a fixed Haar probe set, the fraction of variance of the centered activations z explained by the sampled, jointly orthonormalized spans of matrix coefficients, sector by sector in Casimir order, with exact coefficients verified in a unit-test suite (Appendix C) and a sampling floor near 0.2% of variance. Function-space ablations subtract from z the component linearly predictable from a chosen sector's exact matrix coefficients, an intervention well defined even for sectors at the floor. Activation patching replaces the internal product state by reconstructions from chosen coordinates. Because activations are real, a complex irreducible and its conjugate span one real sector of dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} \mathrm{End}_G(W)$ (Lemma 2; Appendix A collects the background).

3 The learned carrier identifies the global form

Across the ten group forms of Table 1, the encoder concentrates on a small faithful carrier of the task group, deterministically across seeds. The half-integer sector of $\mathrm{SU}(2)$ appears if and only if the observations distinguish $\pm g$, with $\cos(z(g), z(-g)) = -0.95 \pm 0.02$ against $+1.000$ in the quotient condition, so the learned code detects the double cover, a property of the global form invisible to the Lie algebra. When the center is not cyclic, no faithful irreducible representation exists, and the models build the forced direct sum. Where a smaller reducible faithful sum competes against a larger faithful irreducible representation, the sum wins, with the alternative under 1% of variance in every seed.

The selection is causal. Subtracting from z the component linearly predictable from the carrier's matrix coefficients collapses ranking to chance (0.02 against a chance rate of 0.016) and center discrimination to its tie value, while the same ablation on any other sector, or on a random subspace of equal rank, leaves both intact. The pattern is a double dissociation between each group and its quotient, with seed standard deviations below 0.005 on every cell (Table 3). It repeats under the

order-three center of $\mathrm{SU}(3)$, where orbit averaging is an exact identity on quotient models and fatal on faithful ones (Table 4), and for $\mathrm{Spin}(4)$, whose center supports three distinct discriminations, each learned block is necessary for exactly the center flips its central character detects (Table 5). The trained scorer is itself a character function: a single character explains 0.97 ± 0.01 ($\mathrm{SU}(2)$, $\chi_{1/2}$) and 0.88 ± 0.01 ($\mathrm{SO}(3)$, χ_1) of the score variance.

The following bound, proved in Appendix B, explains why ranking never forces a larger carrier.

Lemma 1. *Let V be a faithful finite-dimensional unitary representation of a compact group H with character χ_V. Then $\mathrm{Re}\,\chi_V(g) \le \chi_V(e)$ for all $g \in H$, with equality only at $g = e$. Moreover, for X in the Lie algebra of H, $\mathrm{Re}\,\chi_V(\exp X) = \chi_V(e) - \frac{1}{2}\|dV(X)\|_F^2 + O(\|X\|^4)$, and $X \mapsto \|dV(X)\|_F^2$ is positive definite.*

By Lemma 1, a monotone function of $\mathrm{Re}\,\chi_V(ab\,c^{-1})$ ranks the product above every excluded negative for any faithful V, so optimization selects the carrier on cost. Conversely, a nontrivial kernel forces positive ranking error under explicit sampling assumptions (Lemma 3), and the ablations above realize that necessity block by block.

4 The mechanism is the full coefficient block

The learned code is not merely supported on the carrier's isotypic sector. In every seed, the active subspace of z has exactly the real dimension of the full block, 4 for the quaternionic spinor of $\mathrm{SU}(2)$ and 9 for the real-type vector of $\mathrm{SO}(3)$, as the following lemma prescribes (proof in Appendix B).

Lemma 2. *Let $\rho\colon G \to U(V_\rho)$ be an irreducible unitary representation of a compact group G, and let $M_\rho \subseteq L^2(G)$ be the span of its matrix coefficients $g \mapsto \langle u, \rho(g)v\rangle$. Then $M_\rho \cong V_\rho^* \boxtimes V_\rho$ as a representation of $G \times G$ acting by left and right translation, this representation is irreducible, and every nonzero subspace of M_ρ invariant under both translation actions equals M_ρ. For an irreducible real representation W with commutant $D = \mathrm{End}_G(W)$, the corresponding real coefficient block has dimension $(\dim_{\mathbb{R}} W)^2/\dim_{\mathbb{R}} D$ and is the minimal nonzero two-sided translation-stable real subspace of its sector.*

Two-sided structure. Lemma 2 converts full-block identity into a falsifiable property: a full block supports commuting linear left and right translation actions, a proper subspace of the sector cannot. We fit affine operators M_k^L, M_k^R with $z(kg) \approx M_k^L z(g)$, $z(gk) \approx M_k^R z(g)$, evaluated held out. On the trained $\mathrm{SO}(3)$ code, both actions fit at $R^2 = 0.989 \pm 0.001$ with relative commutator defect $\|[M_a^L, M_b^R]\| = 0.012 \pm 0.001$, matching the exact block. On the explicit 6-dimensional frame $g \mapsto (ge_1, ge_2)$, the left action is exactly linear and the right action fails at $R^2 = 0.638$, since the missing column $ge_3 = (ge_1) \times (ge_2)$ is uncorrelated with the retained ones under Haar measure, and a random subspace of the true block at the frame's dimension fails both actions at $R^2 = 0.67$ (full comparison in Table 6). The fitted operators match the representation itself: the trace of M_k^L equals $\dim(W)\,\chi_W(k)$ with correlation 1.000 across group elements, and its eigenvalue phases match those of $\dim(W)$ copies of $W(k)$ to 0.002 radians for $\mathrm{SU}(2)$.

Causal sufficiency. Replacing the internal product state $h(a, b)$ by its ridge reconstruction from the eight realified spinor coefficients of a counterfactual product $a'b'$ redirects the $\mathrm{SU}(2)$ model to select $a'b'$ with probability 0.996–0.999 in every seed, equal to or better than patching the full activation, while reconstruction from the wrong sector redirects at chance. The output h concentrates $89.9 \pm 1.4\%$ of its variance on the diagonal two-argument sector of its carrier, with $98.2 \pm 0.3\%$ of that sector's energy in the subspace spanned by the matrix product $W(a)W(b)$, so the interaction implements the representation's own composition law.

Cost. Whether two-sided closure is why the block wins is a question about training, so we trained matched models on $\mathrm{SO}(3)$ whose observations are the exact frame (6 dimensions) or the exact full block (9 dimensions), all else equal. At statistically equal accuracy (0.891 ± 0.002 against 0.889 ± 0.003), the block model attains a larger margin (7.29 ± 0.16 against 6.49 ± 0.05) at 17% lower total parameter norm, with every module cheaper (Table 7). The frame model is free to reconstruct the missing column nonlinearly, and evidently does, but the reconstruction costs more than the three saved coordinates. Regularized training therefore selects the full block as the cheapest complete mechanism, storage and composition counted together.

5 Casimir-ordered spectral bias under regression

Ranking requires only one faithful character, so it cannot probe how the architecture allocates capacity across sectors. Regression onto the $\mathrm{SU}(2)$ heat kernel $K_t(g) = \sum_j (2j+1) e^{-t\, j(j+1)} \chi_j(g)$, evaluated at $ab\,c^{-1}$, requires every sector the envelope keeps above the noise. Figure 1(b) reports the fraction of score variance per sector, sequentially orthogonalized in Casimir order, divided by the same fraction for the exact target, so 1 means a sector at its required weight (per-checkpoint values in Table 8).

Three regularities emerge. Each trained score is a truncated character expansion, with recovered sectors near ratio 1 and sectors beyond a sharp frontier at 0, the frontier at higher Casimir for smaller t. The frontier advances in Casimir order during training: at $t = 0.05$, the spin-2 recovery climbs from 0.12 to 1.20 over 70k steps while spin $\frac{5}{2}$ stays at 0. The frontier is set by regularization, not capacity: on the band-limited target $\chi_1 + \frac{1}{2}\chi_2$, weight decay 1.0 prunes the spin-2 component and the abandoned 12.6% of target variance equals the final test error, while weight decay 0.1 recovers the target with profile $(1.00, 1.01, 0.94, 0.95)$ and out-of-support sectors at 0.000. A sector survives when its variance contribution exceeds its norm cost, so the spectral cutoff is a cost boundary of the same kind that selects the carrier in Section 3. The target's central character governs parity as well: the band-limited target is central-even, and the model builds an integer-spin code even on the faithful $\mathrm{SU}(2)$ encoding, with the spinor sector at 1%.

6 A revision, and a selection conjecture

The instruments that produced the selection observation also bound it. On $\mathrm{SO}(5)$, the learned carrier is a mixture, vector $\mathbf{5}$ at 0.68–0.70 of variance plus adjoint $\mathbf{10}$ at 0.21–0.23, stable across five seeds, doubled training, four times the candidates, and reduced weight decay. The adjoint fraction falls only when generalization improves, from 0.23 at held-out accuracy 0.55 to 0.17 at 0.73 under tripled data, and no condition solves the task. The minimal-carrier description therefore fails as stated in one of the ten group forms. Both faithful sectors ranking correctly is consistent with Lemma 1, and two hypotheses remain open: asymptotic purification as the task is solved, or a genuinely mixed minimum-cost mechanism for this group, which the cost result of Section 4 makes coherent. Separating them requires a condition that solves $\mathrm{SO}(5)$.

The structural results suggest a selection principle that we can state but not yet prove.

Conjecture 1. *In the factorized architecture with weight decay, among parameter settings achieving population ranking loss within ε of the optimum, the minimum-norm setting realizes, as $\varepsilon \to 0$, the full real coefficient block of a faithful carrier minimizing the total regularized cost of features, composition, and scoring, with score a monotone function of* $\mathrm{Re}\,\chi_V(ab\,c^{-1})$.

The missing ingredients are a norm lower bound tying any ε-optimal score to the effective dimension of the functions it computes on G, and the exclusion of solutions outside the character algebra, the compact-group analogue of the alternative circuits documented for finite groups Stander et al. [2024], Zhong et al. [2023]. Disentangling carrier selection from the conditioning of the quotient observation, which makes every quotient condition harder than its cover in our experiments, is left as future work.

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A Background: representation theory and harmonic analysis on compact groups

This appendix collects the facts the paper uses, with references in place of proofs. Standard sources are Bröcker and tom Dieck [1985] for compact groups and the Peter–Weyl theorem, Sepanski [2007] for the real Schur lemma and the type classification, Folland [1995] for harmonic analysis, and Fulton and Harris [1991] for the representation theory of the classical groups. Throughout, G is a compact Lie group and all representations are finite-dimensional and continuous.

Haar measure and unitarity. A compact group carries a unique translation-invariant Borel probability measure, its Haar measure, and every expectation over group elements in this paper is taken with respect to it. Averaging an arbitrary inner product over the group makes any representation unitary, and every unitary representation decomposes as a finite orthogonal direct sum of irreducible ones [Bröcker and tom Dieck, 1985, Ch. II]. The set of isomorphism classes of complex irreducible representations is denoted $\widehat{G}$, and for $\rho \in \widehat{G}$ we write $d_\rho = \dim V_\rho$ and $\chi_\rho(g) = \operatorname{tr} \rho(g)$ for its character.

Matrix coefficients and orthogonality. For $\rho \in \widehat{G}$ and vectors $u, v \in V_\rho$, the function $g \mapsto \langle u, \rho(g)v\rangle$ is a matrix coefficient of ρ. Schur's lemma states that a linear map commuting with two irreducible representations is zero or an isomorphism, and that the self-intertwiners of a complex irreducible representation are the scalars [Fulton and Harris, 1991, Ch. I]. Its integrated form is the orthogonality of coefficients: in a unitary basis,

$$\int_G \rho(g)_{ij}\,\overline{\sigma(g)_{kl}}\,dg \;=\; \frac{\delta_{\rho\sigma}\,\delta_{ik}\,\delta_{jl}}{d_\rho}, \qquad \int_G \chi_\rho(g)\,\overline{\chi_\sigma(g)}\,dg \;=\; \delta_{\rho\sigma}$$

for $\rho, \sigma \in \widehat{G}$ [Bröcker and tom Dieck, 1985, Ch. II]. These relations are the foundation of the spectroscopy instrument of Section 2: they make the coefficient spans of distinct irreducible representations orthogonal in $L^2(G)$, so variance fractions per sector are well defined, and our implementation verifies them numerically on the sampled probe sets.

The Peter–Weyl theorem. The matrix coefficients of all $\rho \in \widehat{G}$ span a dense subspace of $L^2(G)$, and as a representation of $G \times G$ acting by left and right translation,

$$L^2(G) \;\cong\; \bigoplus_{\rho \in \widehat{G}} M_\rho, \qquad M_\rho \;\cong\; V_\rho^* \boxtimes V_\rho,$$

an orthogonal Hilbert-space direct sum in which each coefficient space M_ρ has dimension d_ρ^2 [Bröcker and tom Dieck, 1985, Ch. III]. Every function a network computes on G therefore has a canonical spectral decomposition, and the paper's central structural question, which sectors a trained model occupies and whether it occupies them fully, is a question about this decomposition. Lemma 2 records the two-sided irreducibility of M_ρ and its real form.

Real representations and Schur type. The networks compute with real numbers, so the relevant decomposition is that of real representations. For an irreducible real representation W, the real Schur lemma states that the commutant $\mathrm{End}_G(W)$ is a real division algebra, hence isomorphic to $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$ [Sepanski, 2007, Ch. III], and the three cases correspond to the classical trichotomy for a complex irreducible representation V: real type when V admits an invariant real structure, quaternionic type when it admits an invariant quaternionic structure, and complex type when $V \not\cong \bar{V}$. Realification acts on dimensions as follows. A real-type V of dimension d yields $\dim_{\mathbb{R}} W = d$ with commutant $\mathbb{R}$. A complex-type or quaternionic-type V of dimension d yields $\dim_{\mathbb{R}} W = 2d$ with commutant $\mathbb{C}$ or $\mathbb{H}$, and in the complex case V and $\bar{V}$ realify to the same W, which is why a complex irreducible and its conjugate span a single real sector in our measurements. In all three cases, the real coefficient block has dimension

$$\dim_{\mathbb{R}} M_W \;=\; \frac{(\dim_{\mathbb{R}} W)^2}{\dim_{\mathbb{R}} \mathrm{End}_G(W)},$$

which evaluates to d^2, $2d^2$, and d^2 in the real, complex, and quaternionic cases. The three dimensions measured in the paper are instances: 9 for the real-type vector of $\mathrm{SO}(3)$, 18 for the complex-type fundamental pair of $\mathrm{SU}(3)$, and 4 for the quaternionic spinor of $\mathrm{SU}(2)$.

Characters, class functions, and the scoring identity. Characters are conjugation-invariant, and the irreducible characters form an orthonormal basis of the class functions in $L^2(G)$ [Bröcker and tom Dieck, 1985, Ch. II]. Two identities used throughout are $\chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$ and, for unitary ρ,

$$\chi_\rho(ab\,c^{-1}) \;=\; \mathrm{tr}\big(\rho(a)\rho(b)\rho(c)^*\big) \;=\; \big\langle \rho(c),\, \rho(a)\rho(b)\big\rangle_{\mathrm{HS}},$$

so a character evaluated at $ab\,c^{-1}$ is a Hilbert–Schmidt pairing of the candidate's coefficient block against the product of the arguments' blocks. This identity is the algebraic content of the sentence "two-sided linearity makes composition bilinear and scoring a pairing" in Section 4, and it is the form in which the finite-group mechanism of Chughtai et al. [2023] generalizes to the compact case.

Central characters and descent to quotients. By Schur's lemma, the center $Z(G)$ acts in any $\rho \in \widehat{G}$ by a character $\omega_\rho \colon Z(G) \to U(1)$, called the central character. For a closed central subgroup Z, the irreducible representations of the quotient G/Z are exactly those ρ with ω_ρ trivial on Z, and a Z-invariant function on G has spectral support only on such ρ. This is the descent rule of Section 3: the orbit-averaged observation map is Z-invariant, so every sector carrying a nontrivial

central character of Z is forbidden, and the measured floors on those sectors are the empirical form of the rule. One consequence used for $\mathrm{Spin}(4)$ deserves a proof, since it forces the direct-sum carriers of Table 1.

Fact 1. *If a compact group admits a faithful irreducible complex representation, its center is isomorphic to a closed subgroup of $U(1)$. In particular, $\mathrm{Spin}(4)$, with center $\mathbb{Z}_2 \times \mathbb{Z}_2$, has no faithful irreducible representation.*

Proof. Under a faithful irreducible ρ, the center acts by the scalars ω_ρ, and faithfulness makes $z \mapsto \omega_\rho(z)$ injective, so $Z(G)$ embeds in $U(1)$. Every finite subgroup of $U(1)$ is cyclic, and $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not. □

The Casimir element and the heat kernel. Fix a bi-invariant metric on G. The quadratic Casimir element of the universal enveloping algebra acts in each irreducible representation ρ as a scalar $C(\rho) \geq 0$, and the matrix coefficients of ρ are eigenfunctions of the Laplace–Beltrami operator on G with eigenvalue $C(\rho)$ [Sepanski, 2007, Ch. 5]. The Casimir therefore plays the role of a squared frequency, and "Casimir order" in the paper is the compact-group analogue of ordering Fourier modes by frequency. The heat kernel at the identity-based variable g is the class function

$$K_t(g) \;=\; \sum_{\rho \in \widehat{G}} d_\rho \, e^{-t\,C(\rho)} \, \chi_\rho(g),$$

the fundamental solution of the heat equation on G [Folland, 1995, Ch. 1], and its envelope $e^{-t\,C(\rho)}$ is the analytic source of the required sector weights in Section 5.

The groups of the experiments. The irreducible representations of $\mathrm{SU}(2)$ are the spin representations D^j of dimension $2j+1$ for $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, with Casimir $C(D^j) = j(j+1)$, quaternionic type for half-integer j and real type for integer j, and central character $(-1)^{2j}$ at $-1 \in \mathrm{SU}(2)$ [Bröcker and tom Dieck, 1985, Ch. II]. The quotient $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm 1\}$ therefore carries exactly the integer spins, which is the double-cover diagnostic of Section 3. The irreducible representations of $\mathrm{SU}(3)$ are labeled by highest weights (p, q) with dimension $\frac{1}{2}(p+1)(q+1)(p+q+2)$ and central character $\omega^{\,p-q}$ at the central element ωI for ω a primitive cube root of unity, so descent to $\mathrm{PSU}(3)$ requires $p \equiv q \pmod 3$; the sectors tracked in the experiments are the fundamental pair $\mathbf{3} \oplus \bar{\mathbf{3}}$ at $(1,0)$ and $(0,1)$, the adjoint $\mathbf{8}$ at $(1,1)$, and the pairs $\mathbf{6} \oplus \bar{\mathbf{6}}$ and $\mathbf{10} \oplus \bar{\mathbf{10}}$ [Fulton and Harris, 1991, Ch. 13]. The group $\mathrm{Spin}(4)$ is isomorphic to $\mathrm{SU}(2) \times \mathrm{SU}(2)$, its irreducible representations are the outer products (j_L, j_R) with central character $\big((-1)^{2j_L}, (-1)^{2j_R}\big)$ on the center $\mathbb{Z}_2 \times \mathbb{Z}_2$, and its three central quotients are $\mathrm{SO}(4)$ (the diagonal $\mathbb{Z}_2$, retaining $j_L + j_R \in \mathbb{Z}$), $\mathrm{SO}(3) \times \mathrm{SU}(2)$ (one factor's $\mathbb{Z}_2$), and $\mathrm{PSO}(4)$ (the full center, retaining $j_L, j_R \in \mathbb{Z}$). The group $\mathrm{Spin}(5)$ is isomorphic to the quaternionic unitary group $\mathrm{Sp}(2)$, with quaternionic spinor representation $\mathbf{4}$, real vector representation $\mathbf{5}$, and adjoint $\mathbf{10}$, and $\mathrm{SO}(5) = \mathrm{Spin}(5)/\{\pm 1\}$ retains the representations with trivial central character, among them $\mathbf{5}$ and $\mathbf{10}$ [Fulton and Harris, 1991, Ch. 16].

B Proofs

Proof of Lemma 1. Each eigenvalue of the unitary matrix $V(g)$ lies on the unit circle, so $\mathrm{Re}\,\mathrm{tr}\,V(g) \leq \dim V$, with equality exactly when every eigenvalue equals 1, that is, when $V(g)$ is the identity. By faithfulness, this occurs only at $g = e$. For the expansion, write $A = dV(X)$, an anti-Hermitian matrix with eigenvalues $i\lambda_1, \dots, i\lambda_{\dim V}$. Then $\mathrm{Re}\,\mathrm{tr}\exp(tA) = \sum_k \cos(t\lambda_k) = \dim V - \frac{t^2}{2}\sum_k \lambda_k^2 + O(t^4)$, and $\sum_k \lambda_k^2 = \|A\|_F^2$. The trace of A is purely imaginary and contributes nothing to the real part. If $\|dV(X)\|_F = 0$, then the one-parameter subgroup $\exp(tX)$ lies in the kernel of V, which is trivial, so $X = 0$, proving positive definiteness. □

Lemma 3. *Let H be a compact group. Assume the score factors as $s(a,b,c) = S(\varphi(a), \varphi(b), \varphi(c))$ where φ factors through the matrix coefficients of a representation V of H with nontrivial closed kernel N, the function $c \mapsto s(a,b,c)$ is continuous, negatives are sampled from a distribution assigning positive probability to every open subset of the admissible region, and the exclusion radius is smaller than $\min_{n \in N,\, n \neq e} d(e,n)$. Then the population ranking loss is bounded away from its optimum.*

Proof. Fix $n \in N$ with $n \neq e$. By the radius hypothesis, $ab\,n$ lies in the admissible negative region. Since φ is constant on N-cosets, $s(a, b, ab\,n) = s(a, b, ab)$, which is at least the score of the noisy positive up to the continuity modulus of s over the positive displacement. By continuity, there is a ball around $ab\,n$ on which the score stays within any prescribed gap of this value, and by the sampling hypothesis that ball receives positive probability, so with positive probability a sampled negative ties or exceeds the positive. □

The hypotheses of Lemma 3 are necessary: with a finite kernel and a continuous candidate distribution, the exact ambiguous points have measure zero, and without continuity of the score no positive lower bound follows from non-separation alone.

Proof of Lemma 2. The map $V_\rho^* \otimes V_\rho \to L^2(G)$ sending $u^* \otimes v$ to $g \mapsto \langle u, \rho(g)v\rangle$ intertwines the $G \times G$-action by left and right translation with the outer tensor product action and is injective by the orthogonality of matrix coefficients (Appendix A). The outer tensor product of irreducible representations is an irreducible representation of the product group, so M_ρ has no proper nonzero two-sided invariant subspace, and $\dim_{\mathbb{C}} M_\rho = (\dim V_\rho)^2$. For the real form, the real Schur lemma gives $\mathrm{End}_G(W) \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, and the real matrix-coefficient span of W is identified with $W^* \otimes_D W$, of real dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} D$. Minimality follows from irreducibility of the two-sided action on this space, by the same outer-tensor argument applied over D. □

C Experimental details

Groups and sampling. Elements of $\mathrm{SU}(2)$ are unit quaternions, of $\mathrm{SU}(3)$ Haar-random special unitary matrices, of $\mathrm{Spin}(4) = \mathrm{SU}(2) \times \mathrm{SU}(2)$ pairs of unit quaternions, and of $\mathrm{Spin}(5) = \mathrm{Sp}(2)$ quaternionic 2×2 unitaries sampled by quaternionic Gram–Schmidt. Matrix coefficients for half-integer spins of $\mathrm{SU}(2)$ are computed exactly through the action on homogeneous polynomials, for $\mathrm{SU}(3)$ through symmetric powers of the fundamental representation and the adjoint action on the Gell-Mann frame, and for $\mathrm{Sp}(2)$ through the complex embedding of the quaternionic matrices together with the conjugation actions on Hermitian and anti-Hermitian quaternion matrices.

Quotient encodings. Quotient conditions average the observation map over the central subgroup being quotiented: the antipodal average for $\mathrm{SO}(3)$ and $\mathrm{SO}(5)$, the order-three orbit average for $\mathrm{PSU}(3)$, and the three sign-pair averages for the quotients of $\mathrm{Spin}(4)$. Each average is exactly invariant under the quotiented subgroup and distinguishes all other center elements, which we verify numerically at construction. The two further observation maps under which the headline results reproduce are a frozen random two-layer tanh network of the raw coordinates and the values of a fixed random Fourier function at 32 translated probe points, each with its orbit-averaged quotient variant.

Training. Batch size 4096 (2048 for the larger groups), learning rate 10^{-3}, weight decay 1.0, 100k–200k steps, training fraction 0.01 of the pair table unless stated. The ranking results of Table 1 use five seeds per condition. Heat-kernel and band-limited regressions replace the cross-entropy by mean squared error on standardized targets, with candidates drawn half near the product at uniform geodesic radius and half from Haar measure. Under the $\mathrm{SU}(2)$ sampling, the Bayes-optimal ranking rule attains accuracy 0.951.

Analyses. Isotypic fractions use jointly orthonormalized sampled coefficient spans processed in Casimir order, which bounds cross-sector leakage at the sampling floor. Function-space ablations subtract the component of z linearly predictable from a sector's exact matrix coefficients, fitted on an independent probe set. Subspace estimates of near-floor sectors are noise-dominated and ablate incorrectly, which is why the function-space form is the reported instrument. Ridge probes select the regularization strength on a validation split and report on a disjoint test split. Translation operators are fitted by ridge regression in a PCA frame retaining 99.9% of activation variance, with homomorphism and commutator defects evaluated on held-out elements. The score-level spectral profiles regress both the learned score and the exact target on the same sequentially orthogonalized character design, so standardization and collinearity cancel in the reported ratios. Raw per-character least-squares coefficients are unidentifiable under this sampling because high-Casimir characters are near-collinear near the identity, and the sequentially orthogonalized profile is the identifiable replacement.

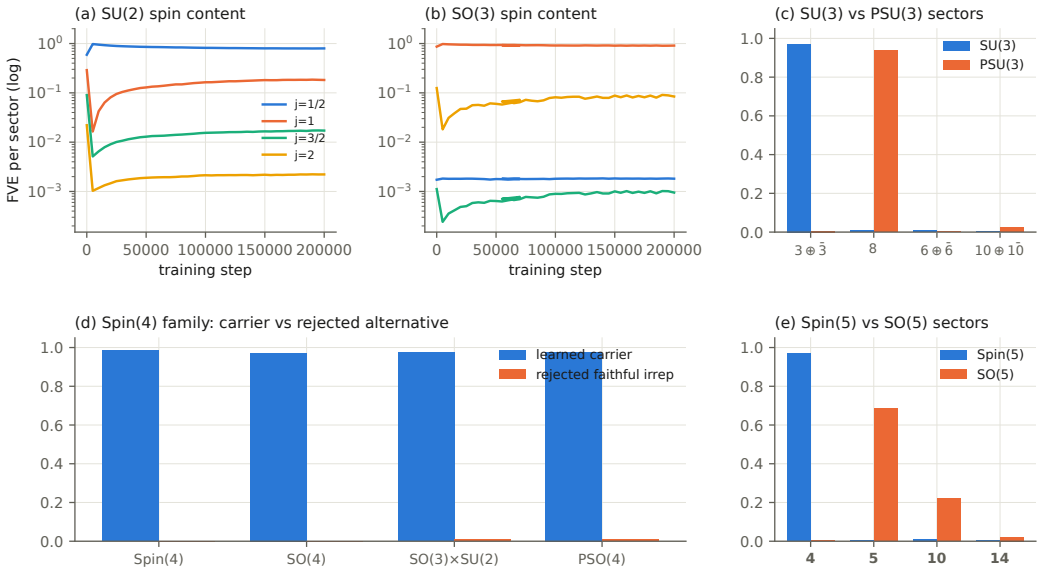


Figure 2: (a, b) Fraction of encoder variance per spin sector during training for the $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ conditions (seed 0, log scale; the dotted floor is the sampling limit). (c, d, e) Final sector spectra for $\mathrm{SU}(3)/\mathrm{PSU}(3)$, the $\mathrm{Spin}(4)$ family, and $\mathrm{Spin}(5)/\mathrm{SO}(5)$ (means over five seeds).

Table 2: Seed statistics for the principal conditions (five seeds, mean $\pm$ standard deviation, evaluated at 100k steps). D denotes the center discrimination appropriate to the condition: the probability that the true product outscores its center translates, with tie value 0.5 ($\mathrm{SU}(2)/\mathrm{SO}(3)$) or $1/3$ ($\mathrm{SU}(3)/\mathrm{PSU}(3)$).

	$\mathrm{SU}(2)$	$\mathrm{SO}(3)$	$\mathrm{SU}(3)$	$\mathrm{PSU}(3)$
held-out accuracy	0.921 ± 0.003	0.883 ± 0.003	0.994 ± 0.001	0.685 ± 0.020
carrier variance	0.835 ± 0.010	0.917 ± 0.008	0.971 ± 0.009	0.934 ± 0.007
forbidden-sector variance	0.0023 ± 0.0004	0.0023 ± 0.0004	0.0059 ± 0.0004	0.0000
center cosine	-0.72 ± 0.02	$+1.000$	-0.484 ± 0.005	$+1.000$
carrier decode R^2	0.995	0.999	0.978 ± 0.006	0.699 ± 0.012
D, intact	1.0000	0.5000	1.0000	0.3333
score–character R^2	0.974 ± 0.006	0.876 ± 0.014	0.700	

Verification. The representation-theoretic primitives pass a unit-test suite covering the homomorphism property, unitarity, characters, central characters, Schur orthogonality, planted-sector recovery, and exact invariance of the quotient encodings. The spectral-recovery analysis reproduces the known support and coefficients of the band-limited target at weight decay 0.1, which serves as the calibration of the instrument. The orbit-average intervention is an exact identity on quotient models by construction and measures as one, which confirms the intervention machinery. Software: PyTorch 2.11 and NumPy on a single T4 GPU. A tracked results bundle records, for every run, the configuration, seed, checkpoint hash, final metrics, and analysis outputs, and every table in this paper regenerates from the bundle. Code and the bundle are included as anonymized supplementary material.

D Complete results

Figure 2 shows the training dynamics of the isotypic spectra and the per-group sector spectra behind Table 1. Table 2 lists the seed statistics for the four principal conditions. Tables 3 to 5 list the intervention results, Table 6 the two-sided translation test, Table 7 the frame-versus-block comparison, and Table 8 the spectral recovery per checkpoint.

Table 3: Interventions on the argument encoders for the $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ conditions (five seeds, mean $\pm$ standard deviation). Task is ranking accuracy against 63 excluded Haar negatives (chance 0.016), D_c the probability that the true product outscores its antipode (tie 0.5).

	$\mathrm{SU}(2)$		$\mathrm{SO}(3)$	
intervention	task	D_c	task	D_c
none	1.000	1.0000 ± 0.0000	1.000	0.5000
ablate spin $\frac{1}{2}$	0.026 ± 0.002	0.503 ± 0.006	1.000 ± 0.000	0.5000
ablate spin 1	1.000 ± 0.000	1.0000	0.027 ± 0.003	0.5000
center-orbit average	0.021	0.498	1.000 (exact identity)	0.5000
random subspace, equal rank	1.000	1.0000	1.000	0.5000

Table 4: Interventions for the $\mathrm{SU}(3)$ and $\mathrm{PSU}(3)$ conditions (five seeds). D_ω is the probability that the true product outscores both of its center translates (tie $1/3$).

	$\mathrm{SU}(3)$		$\mathrm{PSU}(3)$	
intervention	task	D_ω	task	D_ω
none	0.998 ± 0.001	1.000	0.739 ± 0.019	0.333
center-orbit average	0.016 ± 0.001	0.337 ± 0.008	0.739 (exact identity)	0.333
remove invariant part	0.998	1.000	0.015 ± 0.001	0.333
ablate $\mathbf{3} \oplus \bar{\mathbf{3}}$	0.018 ± 0.002	0.335	0.733	0.333
ablate adjoint $\mathbf{8}$	0.997	1.000	0.014 ± 0.003	0.333

Table 5: Block-resolved interventions for the $\mathrm{Spin}(4)$ family (five seeds; representative cells with seed statistics). D_L, D_R, D_{LR} are the discriminations of the three center flips (tie 0.5). Each learned block is necessary for exactly the flips its central character detects.

condition	intervention	task	D_L	D_R
$\mathrm{Spin}(4)$	none	0.997	1.000	1.000
$\mathrm{Spin}(4)$	ablate $(\frac{1}{2}, 0)$	0.57	0.507 ± 0.007	1.000 ± 0.000
$\mathrm{Spin}(4)$	ablate $(0, \frac{1}{2})$	0.57	1.000	0.498
$\mathrm{SO}(4)$	ablate $(\frac{1}{2}, \frac{1}{2})$	0.017 ± 0.001	0.489	0.489
$\mathrm{SO}(3)\times\mathrm{SU}(2)$	ablate $(0, \frac{1}{2})$	0.53	0.500	0.493 ± 0.006
$\mathrm{SO}(3)\times\mathrm{SU}(2)$	ablate $(1, 0)$	0.853 ± 0.036	0.500	1.000
$\mathrm{PSO}(4)$	ablate $(1, 0)$ or $(0, 1)$	0.809 ± 0.016	0.500	0.500

Table 6: The two-sided translation test on the $\mathrm{SO}(3)$ condition (five seeds for the trained and random-initialization systems). Left and right columns are held-out R^2 of fitted affine translation operators, hom-R the right-homomorphism defect, and $[L, R]$ the commutator defect, both relative to activation norm.

feature system	dim	left R^2	right R^2	hom-R	$[L, R]$
trained code	9	0.989 ± 0.001	0.989 ± 0.001	0.012	0.012 ± 0.001
true full block	9	1.000	1.000	0.000	0.000
faithful frame	6	1.000	0.638	0.526	0.039
random truncation of block	6	0.674	0.675	0.411	0.448
raw observations	32	0.992	0.991	0.058	0.073
random-initialization encoder	16	0.892 ± 0.008	0.890 ± 0.008	0.179	0.160 ± 0.014

Table 7: Frame-versus-block matched training on $\mathrm{SO}(3)$ (five seeds per arm, 60k steps). Norms are Frobenius norms of module parameters at the end of training.

features	accuracy	margin	encoder norm	interaction norm	scorer norm	margin/norm
frame, 6 dims	0.891 ± 0.002	6.49 ± 0.05	19.6	11.9	13.7	0.243 ± 0.002
block, 9 dims	0.889 ± 0.003	7.29 ± 0.16	16.7	9.8	12.2	0.319 ± 0.008

Table 8: Score-level spectral recovery per checkpoint (sequentially orthogonalized profile, model divided by target; sectors beyond spin 2 are at 0.000 throughout and omitted). The band-limited target is $\chi_1 + \frac{1}{2}\chi_2$.

target	step	$j = \frac{1}{2}$	$j = 1$	$j = \frac{3}{2}$	$j = 2$
$K_t, t = 0.3$	25k	1.13	1.12	0.13	0.81
$K_t, t = 0.3$	60k	1.05	1.05	1.01	0.07
$K_t, t = 0.1$	60k	1.47	1.39	1.43	0.00
$K_t, t = 0.05$	50k	1.81	1.81	1.71	0.12
$K_t, t = 0.05$	100k	1.63	1.63	1.49	0.83
$K_t, t = 0.05$	120k	1.58	1.55	1.46	1.20
band-limited, wd 1.0	60k	1.01	1.20	0.00	0.00
band-limited, wd 0.1	60k	1.00	1.01	0.94	0.95